

Informe de experimento: results_validation_gpt-4o-mini_agents

Generado el 2026-09-16 10:46 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio results_validation_gpt-4o-mini_agents (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_gpt-4o-mini_agents
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-4o-mini
- Métodos: pfagent
- Casos: case14
- Requests: generadas, N=40 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=10, parameterized=10, multistep=10, ambiguous=10
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 10:41
- Costo total: 0.0667 USD (40 items; costo medio 0.00167 USD/item)
- Items totales: 40; ok=40; con error=0; tokens totales=417643
- Solved global: 95.0% (38/40)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método pfagent

System prompt (SYSTEM_PROMPT_EN) —1375 caracteres; architecture=react, gate=False, memory=True

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

User prompt (primer item) = request_text —67 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

Load case14 and disconnect the line between bus ten and bus eleven.

3. Resultados agregados

Columnas del protocolo (medias por método $\times$ caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfagent_definicion`, **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método $\times$ caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	V pass	\$
pfagent	case14	40	95.0 (38/40)	1.90e-07	0.02	95.0 (38/40)	100.0 (40/40)	0	97.5	–	2.5	10441	–	0.0 (0/40)	97.5 (39/40)	0.0667

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
pfagent	100.0 (10/10)	100.0 (10/10)	90.0 (9/10)	90.0 (9/10)

Solved por dificultad (%, items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
pfagent	100.0 (10/10)	100.0 (10/10)	90.0 (9/10)	90.0 (9/10)

Verificación final (pfagent: $V(x,c,z,y)$ aplicado a la respuesta final)

`pass_first` = pasó al primer intento; `pass_retry` = pasó tras el único reintento con el veredicto añadido; `abstained` = falló dos veces, respuesta forzada a fallo declarado sin números; `retry_mutation` = el reintento intentó una herramienta que muta la red (bloqueada, nunca ejecutada; cuenta como abstención) —ver “Integridad del reintento” abajo. N = items con `final_gate` activo (no todos los métodos lo tienen).

Method	N	pass_first	pass_retry	abstained	retry_mutation
pfagent	40	90.0 (36/40)	2.5 (1/40)	7.5 (3/40)	0.0 (0/40)

Integridad del reintento

De 4 ítem(s) que llegaron a un reintento, 0 intentaron una herramienta que muta la red (bloqueada) en vez de corregir el reporte; se cuentan como abstención, no como “pasó”. El mensaje de reintento ya no sugiere la corrección específica (p. ej. “reconnect...”) y bloquea estructuralmente `modify_load`, `disconnect_line`, `reconnect_line`, `apply_remedial_action` y `load_case` mientras el reintento está en efecto.

Tipos de error de formulación (conteo de items)

Method	extra_step	ok	wrong_id
pfagent	1	38	1

Errores de ejecución por método (conteo de items con error)

(ningún item con *error*)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (`claimed_success_on_failure`, `stale_state`, `abstained`). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método pfagent

Ejemplo: resuelto (solved=True)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.782856812869876, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": ...[5930 chars más]}
gate: PASS (reasons=-, mismatch=7.85e-09 MW)
--- Ronda 2 ---
LLM: The case14 has been loaded, and the line between bus 10 and bus 11 has been successfully disconnected.

- **Key numbers**: Total load: 253.85 MW, Total generation: 266.51 MW, Total losses: 12.66 MW, Voltage range: 1.01 to 1.09 p.u.
- **Violations**: There are multiple overvoltage violations at buses ...[150 chars más]
--- Respuesta final ---
The case14 has been loaded, and the line between bus 10 and bus 11 has been successfully disconnected.

- **Key numbers**: Total load: 253.85 MW, Total generation: 266.51 MW, Total losses: 12.66 MW, Voltage range: 1.01 to 1.09 p.u.
- **Violations**: There are multiple overvoltage violations at buses 1, 6, 7, 8, 11, and 12.
- **Suggested next step**: Consider running a contingency analysis to assess the impact of this disconnection on the network.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=4353;
↳ cost_usd=0.0007; wall=3.0 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-ambiguous-039-s0, case14, difficulty=ambiguous, seed=315597205): Load the 14-bus test case (case14) and set the active load
↳ at bus 1 (0-based) to 11 MW.
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 2, "p_mw": 11.0})
executed_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 1, "p_mw": 11})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.83309137785733,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: modify_load({"bus_id": 1, "p_mw": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.823083955414936, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0099999 ...[6289 chars más]}
gate: PASS (reasons=-, mismatch=6.18e-09 MW)
--- Ronda 3 ---
LLM: The active load at bus 1 has been set to 11 MW, and the power flow has been successfully solved.

- **Total Load**: 263.83 MW
- **Total Generation**: 276.50 MW
- **Total Losses**: 12.67 MW
- **Voltage Range**: 1.035 to 1.090 p.u.

**Violations**: There are multiple overvoltage violations at buses 1, ...[137 chars más]
--- Respuesta final ---
The active load at bus 1 has been set to 11 MW, and the power flow has been successfully solved.

- **Total Load**: 263.83 MW
- **Total Generation**: 276.50 MW
- **Total Losses**: 12.67 MW
- **Voltage Range**: 1.035 to 1.090 p.u.

**Violations**: There are multiple overvoltage violations at buses 1, 6, 7, 8, 9, 10, 11, and 12.

**Suggested Next Step**: Consider running a remedial action analysis to address the overvoltage violations.
--- Scoring ---
formulation=wrong_id (exact=False); V_MAE=7.59e-06; F_MAE=0.61; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=5365; cost_usd=0.0009; wall=3.6 s
```

Ejemplo: no resuelto

```
request (case14-multistep-022-s0, case14, difficulty=multistep, seed=1083806780): Load case14, disconnect the line between bus 9 and bus
↳ 10, set the active load at bus 3 to 118.6 MW, set the active load at bus 4 to 53.6 MW, and finally run an N-1 contingency analysis
↳ ranked by max violations and report the 8 worst outages.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 9, "to_bus": 10}) -> modify_load({"bus_id": 3, "p_mw":
↳ 118.6}) -> modify_load({"bus_id": 4, "p_mw": 53.6}) -> run_n1_contingency({"top_k": 8, "criteria": "max_violations"})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 9, "to_bus": 10}) -> modify_load({"bus_id": 3, "p_mw":
↳ 118.6}) -> modify_load({"bus_id": 4, "p_mw": 53.6}) -> run_n1_contingency({"top_k": 8, "criteria": "max_violations"}) ->
↳ run_powerflow({}) -> run_n1_contingency({"top_k": 8, "criteria": "max_violations"})
fuente de la traza: archivo de traza completo
```

```

--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 264.0857422973543,
  ↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 9, "to_bus": 10})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -5.032452347367612, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[6092 chars más]
gate: PASS (reasons=-, mismatch=6.74e-09 MW)
tool: modify_load({"bus_id": 3, "p_mw": 118.6})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -5.627497640283712, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_p ...[6074 chars más]
gate: PASS (reasons=-, mismatch=7.38e-09 MW)
tool: modify_load({"bus_id": 4, "p_mw": 53.6})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.672176280633835, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0100000 ...[6073 chars más]
gate: PASS (reasons=-, mismatch=7.73e-09 MW)
--- Ronda 3 ---
tool: run_n1_contingency({"top_k": 8, "criteria": "max_violations"})
output: {"plot_type": "n1_ranking", "figure_json":
  ↳ {"data": [{"cliponaxis": false, "hoverinfo": {"text": "\u003cb\u003eOutage\u003c\u002fb\u003e: 10-11",
  ↳ (line)\u003cbr\u003e\u003cb\u003eConverged\u003c\u002fb\u003e: True\u003cbr\u003e\u003cb\u003eVoltage violations\u003c\u003c ...
  ↳ [19612 chars más]
gate: n/a (no es un resultado de flujo)
--- Ronda 4 ---
LLM: The analysis of the IEEE 14-bus system after the specified modifications has been completed.

1. **Summary**: The N-1 contingency analysis identified the worst outages, with significant voltage violations present.
2. **Key Numbers**:
  - Total Load: 264.09 MW
  - Total Generation: 278.14 MW
  - ...[343 chars más]
--- Ronda 4 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
  ↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -5.672176280633835, "is_violation": false,
  ↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0100000 ...[6073 chars más]
gate: PASS (reasons=-, mismatch=7.73e-09 MW)
--- Ronda 5 ---
tool: run_n1_contingency({"top_k": 8, "criteria": "max_violations"})
output: {"plot_type": "n1_ranking", "figure_json":
  ↳ {"data": [{"cliponaxis": false, "hoverinfo": {"text": "\u003cb\u003eOutage\u003c\u002fb\u003e: 10-11",
  ↳ (line)\u003cbr\u003e\u003cb\u003eConverged\u003c\u002fb\u003e: True\u003cbr\u003e\u003cb\u003eVoltage violations\u003c\u003c ...
  ↳ [19612 chars más]
gate: n/a (no es un resultado de flujo)
--- Ronda 6 ---
LLM: The analysis of the IEEE 14-bus system after the specified modifications has been completed and verified.

1. **Summary**: The N-1 contingency analysis identified the worst outages, with significant voltage violations present.
2. **Key Numbers**:
  - Total Load: 286.75 MW
  - Total Generation: 30 ...[356 chars más]
--- Respuesta final ---
The analysis of the IEEE 14-bus system after the specified modifications has been completed and verified.

1. **Summary**: The N-1 contingency analysis identified the worst outages, with significant voltage violations present.
2. **Key Numbers**:
  - Total Load: 286.75 MW
  - Total Generation: 304.21 MW
  - Total Losses: 17.46 MW
  - Voltage Range: 1.01 to 1.09 p.u.
3. **Violations**: There are multiple voltage violations at several buses, particularly with bus 1, which has 8 violations.
4. **Suggested Next Step**: Consider implementing remedial actions to address the voltage violations or further analyze the system for
  ↳ potential improvements.
--- Scoring ---
formulation=extra_step (exact=False); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=False (formulation); flags=-; error=-; tokens=94131;
  ↳ cost_usd=0.0143; wall=11.6 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- Todos los métodos que llegan al solver (pfagent) tienen $V_MAE < 1e-4$ p.u. en todos los casos (máximo 1.90e-07 en pfagent/case14).
- El gate verificó 46 resultados de flujo, 0 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved de pfagent: 95.0 % (38/40).
- Formulación exacta agregada sobre métodos con tools: 95.0 % (38/40).
- Formulación por dificultad: mejor plain 100.0 % (10/10), peor multistep 90.0 % (9/10).

- Errores de formulación más frecuentes: `extra_step=1`, `wrong_id=1`.
- Ningún item reclamó éxito sobre un fallo (`claimed_success_on_failure=0`).
- 0 de 26 items con mutación de red reportaron estado obsoleto (`stale_state`).
- Trazabilidad de números (Faith.) media con tools: 97.5 %.
- Costo total 0.0667 USD; el método más caro fue `pfagent` con 0.0667 USD (100 %).
- Métodos con tools: 2.50 tool calls medias por item (máximo 7).

6. Archivos

Rutas absolutas bajo `results_validation_gpt-4o-mini_agents.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_validation_gpt-4o-mini_agents/report.json    [leído:
↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_agents/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_agents/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_agents/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_agents/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_agents/scoreboard.rescored.json
scoreboard_per_case.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_agents/scoreboard_per_case.rescored.json
traces (40 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-4o-mini_agents/traces
```